THE NEGATIVE PROLATE SPECTRUM AND THE RIEMANN–SIEGEL THETA FUNCTION

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Abstract. Connes and Moscovici found that the negative eigenvalues of the self-adjoint prolate wave operator reproduce the growth of the zeros of the Riemann zeta function. In the published version, their counting law for the associated Dirac operator contains a term $-(\log E)/2\pi$. We prove that this term is absent. The proof is an elementary phase analysis, and it places the eigenvalues, up to $O(t^{-2/3})$, at the points where the Riemann–Siegel theta function equals $(n-\frac{3}{8})\pi$.

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1. Introduction

For $\lambda > 0$ the prolate wave operator is the differential expression

$$(W\xi)(x) = -\frac{\mathrm{d}}{\mathrm{d}x}\Big((\lambda^2 - x^2)\frac{\mathrm{d}\xi}{\mathrm{d}x}\Big) + (2\pi\lambda x)^2\,\xi(x), \qquad x \in \mathbb{R}. \tag{1}$$

On the interval $[-\lambda, \lambda]$ it commutes with time and band limiting, which is the observation of Slepian and Pollak behind the prolate spheroidal wave functions [10]. Connes and Moscovici studied W on the whole line [6, 7]. They showed that W has a unique self-adjoint extension W_{sa} in $L^2(\mathbb{R})$ that commutes with the projection P_λ onto functions supported in $[-\lambda, \lambda]$ and with its Fourier conjugate. The spectrum of W_{sa} is discrete and unbounded in both directions. The positive part on $[-\lambda, \lambda]$ is the classical prolate spectrum. The negative eigenvalues live outside the interval, their eigenfunctions lie in the Sonin space, and their growth matches that of the squares of the zeros of $\zeta(\frac{1}{2} + it)$. This last point ties the operator to the positivity of Weil's functional at the archimedean place [5].

For $\lambda = \sqrt{2}$, Connes and Moscovici built a Dirac-type square root $2\not{D}$ of the even part of W_{sa} outside the interval, using a Darboux factorisation. Its imaginary eigenvalues are $\pm 2\sqrt{\mu}$, where μ runs over the negative eigenvalues. Let $N(E)$ be the number of these eigenvalues with imaginary part in $(0, E]$. The preprint [6, Thm. 6.1] states

$$N(E) = \frac{E}{2\pi}\Big(\log\frac{E}{2\pi} - 1\Big) + O(1), \tag{2}$$

which is the smooth part of the Riemann–von Mangoldt formula. Its proof, however, rests on a semiclassical phase-space count [6, Prop. 4.2] that is not a proof about eigenvalues. The published version [7, Thm. 5.1, Eq. (33)] instead asserts

$$N(E) = \frac{E}{2\pi}\Big(\log\frac{E}{2\pi} - 1\Big) - \frac{\log E}{2\pi} + O(1), \tag{3}$$

and derives it from Heywood's formula [8] and later extensions of it, applied after a Liouville transformation. The extra term would push the spectrum away from the zeros by an unbounded amount, against the paper's own figures.

Formula (3) is false. The hypotheses of Heywood's formula fail, as also observed in [11], at the endpoint $x = \lambda$, where the reduced potential has the critical singularity $-1/(4t^2)$. There the correct contribution is a fixed Bessel phase $\pi/4$, not a logarithmically growing term. We prove the law (2) for every $\lambda > 0$ and both parities, with a quantisation rule for the individual eigenvalues. Write

$$\kappa = 2\pi\lambda^2, \qquad \Phi(E) = E\log\frac{4E}{\kappa} - E + \frac{\pi}{4}. \tag{4}$$

Theorem A. *Let W_{sa}^{+} and W_{sa}^{-} be the restrictions of W_{sa} to even and to odd functions. There is a continuous, strictly increasing function $\Theta : [0,\infty) \to \mathbb{R}$ with*

$$\Theta(E) = \Phi(E) + O(E^{-2/3}) \qquad (E \to \infty)$$

such that, for $E > 0$, the number $-E^2$ is an eigenvalue of W_{sa}^{+} exactly when $\Theta(E) \in \pi\mathbb{Z}$, and of W_{sa}^{-} exactly when $\Theta(E) \in \frac{\pi}{2} + \pi\mathbb{Z}$. All these eigenvalues are simple. Consequently the number $N_{\pm}(E)$ of eigenvalues of $W_{\mathrm{sa}}^{\pm}$ in $[-E^2, 0)$ satisfies

$$N_{\pm}(E) = \frac{E}{\pi}\Big(\log\frac{2E}{\pi\lambda^2} - 1\Big) + O(1),$$

and there are integers $n_{\pm}$ such that the eigenvalues $-(E_n^{\pm})^2$, listed with $E_1^{\pm} < E_2^{\pm} < \cdots$, satisfy

$$E_n^{\pm}\log\frac{2E_n^{\pm}}{\pi\lambda^2} - E_n^{\pm} = \Big(n + n_{\pm} \mp \frac{1}{4}\Big)\pi + O\big((E_n^{\pm})^{-2/3}\big).$$

Thus the even and the odd negative eigenvalues interlace, and both are governed by a single phase function. Figure 1 shows the levels.

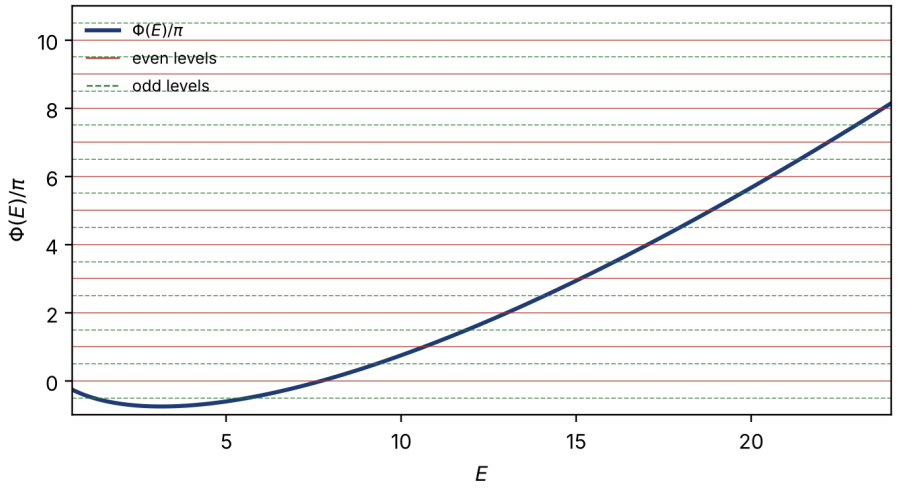


FIGURE 1. The function $\Phi(E)/\pi$ for $\lambda = \sqrt{2}$. Up to $O(E^{-2/3})$, even negative eigenvalues $-E^2$ occur where it meets an integer, odd ones where it meets a half-integer.

For $\lambda = \sqrt{2}$ the phase Φ turns out to be a shifted copy of the Riemann–Siegel theta function

$$\vartheta(t) = \arg\Gamma\Big(\frac{1}{4} + \frac{it}{2}\Big) - \frac{t}{2}\log\pi = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O\Big(\frac{1}{t}\Big).$$

Theorem B. *Let $\lambda = \sqrt{2}$ and let $0 < t_1 < t_2 < \cdots$ be the positive imaginary parts of the eigenvalues of $2\mathbb{D}$. There is an integer n_0 with*

$$\vartheta(t_n) = \Big(n + n_0 - \frac{3}{8}\Big)\pi + O\big(t_n^{-2/3}\big).$$

The counting function satisfies $N(E) = \vartheta(E)/\pi + O(1)$, so (2) holds and (3) does not. Moreover

$$N(T) - N_\zeta(T) = -S(T) + O(1),$$

where $N_\zeta(T)$ counts the zeros of $\zeta(s)$ with $0 < \Im s \le T$ and $S(T) = \pi^{-1}\arg\zeta(\frac12 + iT)$. In particular $N(T) - N_\zeta(T)$ is unbounded above and below.

The points $\vartheta(t) = m\pi$ are the Gram points. Theorem B says that the Connes–Moscovici spectrum sits, asymptotically, on the lattice $\vartheta \equiv -\frac{3}{8}\pi$ modulo π, a rigid version of Gram's law. The zeros follow the same lattice on average, $N_\zeta(T) = \vartheta(T)/\pi + 1 + S(T)$, but they fluctuate around it by $S(T)$, and the operator does not see this fluctuation.

Relation to other work. Ramis, Richard-Jung and Thomann proved that the eigenvalues of W_{sa} not coming from the interval problem are negative, and they characterised them through spectral determinants [9]. Taira, Willems and Wrochna have recently obtained the logarithmic Weyl law and a Bohr–Sommerfeld rule for W_{sa} by microlocal methods [11], for every value of the parameter [11, Remark 1.4]. They point out that classical Sturm–Liouville asymptotics such as Heywood's do not apply at the degenerate endpoints. Their quantisation rule is stated in terms of a renormalised action integral, and its remainder is sharper than ours; their Weyl law already rules out (3), although the consequence for the Dirac counting function is not drawn there. The present argument is different and short. It uses only a Prüfer angle, a comparison with the Bessel function J_0 at the singular endpoint, and an integration by parts at infinity. What is new here is the closed form of the phase in terms of ϑ, the resulting Gram-type rule and the comparison $N - N_\zeta = -S + O(1)$, and the explanation of where the term in (3) comes from. This note continues the author's interest in operator-theoretic and noncommutative structures [2, 3, 4].

2. The outer problem

Throughout, $\lambda > 0$ is fixed and $\kappa = 2\pi\lambda^2$. We use the description of the domain of W_{sa} given in [6, §2]. A function ξ in the maximal domain belongs to the domain of W_{sa} if and only if

$$\lim_{x\to\pm\lambda} (\lambda^2 - x^2)\,\xi'(x) = 0 \tag{5}$$

on both sides of each point $\pm\lambda$, and at infinity the even part ξ^+ and the odd part ξ^- satisfy

$$\lim_{x\to\pm\infty} \Big(x\sin(2\pi\lambda x)\,\xi^{+\prime}(x) - \big(2\pi\lambda x\cos(2\pi\lambda x) - \sin(2\pi\lambda x)\big)\xi^+(x)\Big) = 0, \tag{6}$$

$$\lim_{x\to\pm\infty} \Big(x\cos(2\pi\lambda x)\,\xi^{-\prime}(x) + \big(2\pi\lambda x\sin(2\pi\lambda x) + \cos(2\pi\lambda x)\big)\xi^-(x)\Big) = 0. \tag{7}$$

On $(-\lambda,\lambda)$ the operator is nonnegative. Indeed, for ξ in the domain, the condition (5) removes the logarithmic solution at $\pm\lambda$, so ξ is bounded there and $(\lambda^2 - x^2)\xi'\bar\xi \to 0$. An integration by parts gives

$$\int_{-\lambda}^{\lambda} W\xi\,\bar\xi\,\mathrm{d}x = \int_{-\lambda}^{\lambda} \Big((\lambda^2 - x^2)|\xi'|^2 + (2\pi\lambda x)^2|\xi|^2\Big)\,\mathrm{d}x \;\ge 0.$$

Since W_{sa} commutes with P_λ, an eigenfunction for a negative eigenvalue vanishes on $(-\lambda,\lambda)$. Being even or odd, it is determined by its restriction to (λ,∞).

For $E \geq 0$ consider on (λ, ∞) the equation $W\xi = -E^2\xi$, that is,

$$\big((x^2-\lambda^2)u'\big)' + (4\pi^2\lambda^2x^2 + E^2)u = 0. \tag{8}$$

The point $x = \lambda$ is a regular singular point with the double indicial root 0. Hence (8) has a solution u_E that is analytic at λ with $u_E(\lambda) = 1$, and every solution independent of it behaves like $\log(x-\lambda)$. For the latter $(x^2-\lambda^2)u' \to 2\lambda \neq 0$, so (5) selects the multiples of u_E. The Frobenius coefficients of u_E are polynomials in E^2, so u_E and u_E' depend analytically on E^2, locally uniformly in $x \in [\lambda, \infty)$. For C^1 functions f, g on (λ, ∞) we write

$$[f,g](x) = (x^2-\lambda^2)(f(x)g'(x) - f'(x)g(x)), \qquad s(x) = \frac{\sin 2\pi\lambda x}{x}, \qquad c(x) = \frac{\cos 2\pi\lambda x}{x}.$$

Lemma 2.1. *Let $E > 0$. The number $-E^2$ is an eigenvalue of W_{sa}^+ if and only if $[u_E, s](x) \to 0$ as $x \to \infty$, and of W_{sa}^- if and only if $[u_E, c](x) \to 0$. In both cases the eigenvalue is simple.*

Proof. We use from Lemma 3.2 below that every solution f of (8) satisfies $f(x) = O(1/x)$ and $f'(x) = O(1/x)$. Then for such f

$$x\sin(2\pi\lambda x)f' - \big(2\pi\lambda x\cos(2\pi\lambda x) - \sin(2\pi\lambda x)\big)f = -[f,s] - \lambda^2(fs' - f's),$$

and the last term tends to 0. So (6) for an even function equal to f on (λ, ∞) means $[f, s] \to 0$. In the same way (7) means $[f, c] \to 0$.

If $W_{\mathrm{sa}}^+\xi = -E^2\xi$, then by the remarks above ξ vanishes on $(-\lambda, \lambda)$, and on (λ, ∞) it is a multiple of u_E, which gives the condition and the simplicity. Conversely, suppose $[u_E, s] \to 0$ and let $\xi(x) = u_E(|x|)$ for $|x| > \lambda$ and $\xi = 0$ on $[-\lambda, \lambda]$. Then $\xi \in L^2$ by the decay of u_E. For a test function φ, integrating by parts on each of the three intervals produces boundary terms at $\pm\lambda$ of the form $(\lambda^2 - x^2)(\xi\varphi' - \xi'\varphi)$, and these vanish because ξ is bounded and $(\lambda^2 - x^2)\xi' \to 0$. Hence $W\xi = -E^2\xi$ as distributions. The function ξ satisfies (5) and (6), so it lies in the domain of W_{sa}. The odd case is the same with $\xi(x) = \operatorname{sign}(x)\, u_E(|x|)$. □

3. A Prüfer angle

Put $x = \lambda\cosh t$ with $t \in (0, \infty)$. Then $x^2 - \lambda^2 = \lambda^2\sinh^2 t$, $4\pi^2\lambda^2x^2 = \kappa^2\cosh^2 t$, and $(x^2-\lambda^2)\,\mathrm{d}/\,\mathrm{d}x = \lambda\sinh t\,\mathrm{d}/\,\mathrm{d}t$. Writing $\dot f = \mathrm{d}f/\,\mathrm{d}t$, equation (8) becomes

$$\big(\sinh t\,\dot u\big)^{\cdot} + \sinh t\; w(t)\, u = 0, \qquad w(t) = E^2 + \kappa^2\cosh^2 t. \tag{9}$$

Set

$$k(t) = \sqrt{w(t)}, \qquad m(t) = (\lambda\sinh t)^{1/2} w(t)^{1/4}, \qquad a(t) = \frac{\dot m}{m} = \frac{1}{2}\coth t + \frac{\kappa^2\sinh t\cosh t}{2w(t)}.$$

For a real C^1 function f on (λ, ∞) for which f and f' have no common zero, define $\rho > 0$ and a continuous angle φ by

$$mf = \rho\sin\varphi, \qquad \frac{\lambda\sinh t\,\dot f}{m} = \rho\cos\varphi. \tag{10}$$

The angle is determined up to a constant multiple of 2π. Since $\lambda\sinh t\,\dot f = (x^2-\lambda^2)f'$, a direct computation gives, for any two such functions,

$$[f,g] = \rho_f\rho_g\sin(\varphi_f - \varphi_g). \tag{11}$$

If f solves (9), put $y = mf$ and $z = \lambda\sinh t\,\dot f/m$. Using $m^2 = \lambda\sinh t\,k$ and (9) one finds $\dot y = ay + kz$ and $\dot z = -az - ky$, hence

$$\dot\varphi = k + a\sin 2\varphi, \qquad \frac{\dot\rho}{\rho} = -a\cos 2\varphi. \tag{12}$$

These equations are exact. We record two bounds valid for all $t > 0$. Since $0 \le \coth t - 1/t \le \min(t/3, 1)$ and $\kappa^2 \sinh t \cosh t \le w \tanh t$,

$$0 < a(t) \le \frac{1}{2t} + 1, \qquad |\dot a(t)| \le \frac{1}{2\sinh^2 t} + 1. \tag{13}$$

For the second bound, the derivative of $\kappa^2 \sinh t \cosh t/(2w)$ equals $\kappa^2(\cosh^2 t + \sinh^2 t)/(2w) - \kappa^4 \sinh^2 t \cosh^2 t/w^2$, and both terms lie in $[0, 1]$ because $\kappa^2 \cosh^2 t \le w$.

3.1. **The singular endpoint.** Let φ be the angle of $u = u_E$. Since u is analytic in x at λ, we have $\dot u = \lambda \sinh t\, u'(x) \to 0$ as $t \to 0$, while $u \to 1$. From (10), $\cot\varphi = \dot u/(ku) \to 0$, and we fix the angle by $\varphi(0^+) = \pi/2$. Let

$$K(t) = \int_0^t k(\tau)\,\mathrm{d}\tau, \qquad \eta(t) = \varphi(t) - \frac{\pi}{2} - K(t).$$

Lemma 3.1. *$\eta(t)/K(t) \to -\frac12$ as $t \to 0$.*

Proof. Integrating (9) from 0, where $\sinh t\,\dot u \to 0$, gives $\sinh t\,\dot u(t) = -\int_0^t \sinh\tau\, w(\tau)u(\tau)\,\mathrm{d}\tau = -\frac12 w(0)t^2(1+o(1))$. Hence $\cot\varphi = \dot u/(ku) = -\frac12\sqrt{w(0)}\,t\,(1+o(1))$, and so $\varphi - \frac{\pi}{2} = \frac12\sqrt{w(0)}\,t\,(1+o(1))$. Since $K(t) = \sqrt{w(0)}\,t\,(1+o(1))$, the claim follows. □

3.2. **Infinity.** Let $t_1 = \max\{1, \operatorname{arccosh}(3/\kappa)\}$. For $t \ge t_1$ we have $a \le \frac12\coth 1 + \frac12 < \frac65$, $|\dot a| < \frac32$ and $k \ge \kappa\cosh t \ge 3$, so by (12) $\dot\varphi \ge k - a \ge k/2$ for every solution. These bounds do not depend on E.

Lemma 3.2. *There is an absolute constant C such that for every nonzero real solution f of (8), every $E \ge 0$ and every $t \ge t_1$,*

$$\Big|\int_t^\infty a\sin 2\varphi_f\,\mathrm{d}\tau\Big| + \Big|\int_t^\infty a\cos 2\varphi_f\,\mathrm{d}\tau\Big| \le \frac{C}{\kappa\cosh t}.$$

Consequently $\rho_f(t)$ has a positive finite limit, $\varphi_f(t) - \kappa\cosh t$ has a finite limit, and $f(x) = O(1/x)$, $f'(x) = O(1/x)$ as $x \to \infty$.

Proof. Write $\varphi = \varphi_f$. Integration by parts gives

$$\int_t^T a\sin 2\varphi\,\mathrm{d}\tau = \Big[-\frac{a\cos 2\varphi}{2\dot\varphi}\Big]_t^T + \frac12\int_t^T \Big(\frac{a}{\dot\varphi}\Big)^{\cdot}\cos 2\varphi\,\mathrm{d}\tau.$$

The boundary terms are at most $a/k \le 2/(\kappa\cosh\tau)$ at $\tau = t, T$. By (12), $\ddot\varphi = \dot k + \dot a\sin 2\varphi + 2a\dot\varphi\cos 2\varphi$, so

$$\Big|\Big(\frac{a}{\dot\varphi}\Big)^{\cdot}\Big| \le \frac{|\dot a|}{\dot\varphi} + \frac{a\dot k}{\dot\varphi^2} + \frac{a|\dot a|}{\dot\varphi^2} + \frac{2a^2}{\dot\varphi} \le \frac{12}{k} + \frac{5\dot k}{k^2},$$

where we used $\dot\varphi \ge k/2$, $k \ge 3$ and the bounds on $a, \dot a$. Now $\dot k \ge 0$, $\int_t^\infty \dot k/k^2 = 1/k(t)$ and $\int_t^\infty \mathrm{d}\tau/k \le \int_t^\infty \mathrm{d}\tau/(\kappa\cosh\tau) \le 4/(\kappa\cosh t)$. Letting $T \to \infty$ gives the bound for the sine integral, and the cosine integral is treated identically. By (12), $\log\rho_f$ and $\varphi_f - K$ therefore converge. Moreover

$$K(t) - \kappa\cosh t = \int_0^t (k - \kappa\cosh\tau)\,\mathrm{d}\tau - \kappa e^{-t}, \qquad 0 \le k - \kappa\cosh\tau = \frac{E^2}{k + \kappa\cosh\tau} \le \frac{E^2}{2\kappa\cosh\tau},$$

so $K(t) - \kappa\cosh t$ converges, and $\varphi_f - \kappa\cosh t$ converges as well. Finally $|f| \le \rho/m$ and $|(x^2 - \lambda^2)f'| \le \rho m$, with $m \asymp x$ and ρ bounded, which gives the decay. □

By the proof, the limit

$$I(E) := \lim_{t\to\infty} \big(K(t) - \kappa\cosh t\big) = \int_0^\infty \Big(\sqrt{E^2 + \kappa^2\cosh^2 t} - \kappa\cosh t\Big)\,\mathrm{d}t \tag{14}$$

exists, and we define

$$\Theta(E) := \lim_{t\to\infty} \big(\varphi(t) - \kappa\cosh t\big) = \frac{\pi}{2} + I(E) + \eta(\infty), \tag{15}$$

where φ is the angle of u_E normalised by $\varphi(0^+) = \pi/2$. The tail bound of Lemma 3.2 is uniform in E, the convergence in (14) is locally uniform in E, and $\varphi(t_1)$ depends continuously on E because u_E does. Hence Θ is continuous on $[0,\infty)$.

Lemma 3.3. *Let φ_s, ρ_s and φ_c, ρ_c be the data (10) of s and c, with the same m. Then $\rho_s, \rho_c \to \sqrt{2\pi\lambda}$, while $\varphi_s - \kappa\cosh t \to 0$ and $\varphi_c - \kappa\cosh t \to \frac{\pi}{2}$ modulo 2π.*

Proof. In the variable x, $m^2 = \sqrt{(x^2-\lambda^2)(E^2+4\pi^2\lambda^2x^2)} = 2\pi\lambda x^2(1+O(x^{-2}))$. Hence $ms = \sqrt{2\pi\lambda}\,\sin(2\pi\lambda x) + O(x^{-2})$ and, since $(x^2-\lambda^2)s' = 2\pi\lambda x\cos(2\pi\lambda x) - \sin(2\pi\lambda x) + O(1/x)$, also $(x^2-\lambda^2)s'/m = \sqrt{2\pi\lambda}\cos(2\pi\lambda x) + O(1/x)$. Thus $\rho_s e^{i\varphi_s} = \sqrt{2\pi\lambda}\,e^{2\pi i\lambda x} + O(1/x)$, and $2\pi\lambda x = \kappa\cosh t$. For c one gets $\rho_c e^{i\varphi_c} = \sqrt{2\pi\lambda}\,ie^{2\pi i\lambda x} + O(1/x)$. □

Proposition 3.4. *Let $E > 0$. Then $-E^2$ is an eigenvalue of W_{sa}^+ if and only if $\sin\Theta(E) = 0$, and of W_{sa}^- if and only if $\cos\Theta(E) = 0$.*

Proof. By (11) and Lemmas 3.2 and 3.3,

$$[u_E, s] = \rho\rho_s\sin(\varphi-\varphi_s) \longrightarrow \rho(\infty)\sqrt{2\pi\lambda}\,\sin\Theta(E), \qquad [u_E, c] \longrightarrow -\rho(\infty)\sqrt{2\pi\lambda}\,\cos\Theta(E),$$

with $\rho(\infty) > 0$. Lemma 2.1 completes the proof. □

Proposition 3.5. *Θ is strictly increasing on $[0,\infty)$.*

Proof. We compare φ with the angle ψ defined by (10) with m replaced by $M = (\lambda\sinh t)^{1/2}(\kappa\cosh t)^{1/2}$, which does not depend on E, and normalised by $\psi(0^+) = \pi/2$. Then $\tan\varphi = r\tan\psi$ with $r = m^2/M^2 = \sqrt{w}/(\kappa\cosh t) \geq 1$. Both angles meet the lattice $\frac{\pi}{2}\mathbb{Z}$ exactly at the zeros of u and of $\dot u$, and only increasingly: there $\dot\varphi = k > 0$, while the equation for ψ obtained as in (12) gives $\dot\psi = \kappa\cosh t$ at the zeros of u and $\dot\psi = w/(\kappa\cosh t)$ at the zeros of $\dot u$. At $t = 0^+$ both angles exceed $\frac{\pi}{2}$, since $\cot\varphi$ and $\cot\psi$ are negative there by the proof of Lemma 3.1. Therefore $\varphi(t)$ and $\psi(t)$ always lie in the same interval $[j\frac{\pi}{2}, (j+1)\frac{\pi}{2})$. On such an interval, if $\tan\alpha = r\tan\beta$ then $|\alpha-\beta| \leq \frac{1}{2}\log r$, because $\ell \mapsto \arctan(e^\ell T)$ has derivative of modulus at most $\frac{1}{2}$. Since $r \to 1$ as $t\to\infty$, we get $\Theta(E) = \lim_{t\to\infty}(\psi(t) - \kappa\cosh t)$.

Let $\sigma = E^2$ and $v = \partial_\sigma u$. Differentiating (8) gives $\big((x^2-\lambda^2)v'\big)' + (4\pi^2\lambda^2x^2+\sigma)v = -u$, and $v(\lambda) = 0$ with v analytic at λ. Hence $[u,v]' = -u^2$ and $[u,v](\lambda^+) = 0$, so $[u,v](x) = -\int_\lambda^x u^2$. With R the radius attached to ψ, a short computation from (10) gives

$$\partial_\sigma\psi = -\frac{[u,v]}{R^2} = \frac{1}{R^2}\int_\lambda^x u^2\,\mathrm{d}x' > 0.$$

Fix $0 \leq E_1 < E_2$. Since $R^2 \leq r\rho^2$, with $r \leq (1 + E_2^2/\kappa^2)^{1/2}$, and ρ is bounded on $[t_1,\infty)$ uniformly for $E \in [E_1, E_2]$ by Lemma 3.2, there is $c_0 > 0$ with $\partial_\sigma\psi(t) \geq c_0$ for all $t \geq t_1$ and $\sigma \in [E_1^2, E_2^2]$. Integrating in σ and letting $t\to\infty$ yields $\Theta(E_2) - \Theta(E_1) \geq c_0(E_2^2 - E_1^2) > 0$. □

Corollary 3.6. *For $E > 0$,*

$$N_+(E) = \#\{j\in\mathbb{Z} :\ \Theta(0) < j\pi \leq \Theta(E)\}, \qquad N_-(E) = \#\{j\in\mathbb{Z} :\ \Theta(0) < (j+\tfrac{1}{2})\pi \leq \Theta(E)\}.$$

In particular $|N_\pm(E) - \Theta(E)/\pi| \leq 1 + |\Theta(0)|/\pi$.

4. The phase at high energy

Lemma 4.1. *$I(E) = E\log\frac{4E}{\kappa} - E + O(E^{-1}\log E)$ as $E\to\infty$.*

Proof. Let $b = \kappa/2$ and $I_0(E) = \int_0^\infty \bigl(\sqrt{E^2+b^2e^{2t}} - be^t\bigr)\,\mathrm{d}t$. With $v = be^t/E$, and since $\frac{\mathrm{d}}{\mathrm{d}v}\bigl(\sqrt{1+v^2} - v - \operatorname{arcsinh}(1/v)\bigr) = (\sqrt{1+v^2}-v)/v$, we get

$$I_0(E) = E\Bigl(\operatorname{arcsinh}\frac{E}{b} + \frac{b}{E} - \sqrt{1+\frac{b^2}{E^2}}\Bigr) = E\log\frac{4E}{\kappa} - E + b + O(E^{-1}).$$

Since $\kappa\cosh t - be^t = be^{-t}$ and $\kappa^2\cosh^2 t - b^2e^{2t} = \kappa^2(\frac12 + \frac14 e^{-2t}) \in [\frac{\kappa^2}{2}, \frac{3\kappa^2}{4}]$,

$$I(E) - I_0(E) = -b + \int_0^\infty \frac{\kappa^2\cosh^2 t - b^2e^{2t}}{\sqrt{E^2+\kappa^2\cosh^2 t} + \sqrt{E^2+b^2e^{2t}}}\,\mathrm{d}t.$$

The integral lies between 0 and $\frac{3\kappa^2}{8}\int_0^\infty \frac{\mathrm{d}t}{\sqrt{E^2+b^2e^{2t}}} = \frac{3\kappa^2}{8E}\operatorname{arcsinh}\frac{E}{b}$. □

Next we need the phase of J_0. Define β on $(0,\infty)$ by $\sqrt{s}\,J_0(s) = \varrho\sin\beta$, $-\sqrt{s}\,J_1(s) = \varrho\cos\beta$ with $\varrho>0$, β continuous and $\beta(0^+) = \frac{\pi}{2}$, and put $H(s) = \beta(s) - \frac{\pi}{2} - s$. This is the angle (10) for the equation $f'' + f'/s + f = 0$ with $m = \sqrt{s}$, so

$$H'(s) = -\frac{1}{2s}\sin\bigl(2s + 2H(s)\bigr), \qquad \frac{H(s)}{s} \to -\frac12 \quad (s\to 0). \tag{16}$$

The limit follows from $\cot\beta = -J_1(s)/J_0(s) = -\frac{s}{2} + O(s^3)$.

Lemma 4.2. *$H(s) = -\frac{\pi}{4} + O(1/s)$ as $s\to\infty$.*

Proof. By Hankel's expansions, $J_0(s) = \sqrt{2/\pi s}\,\bigl(\cos(s-\frac{\pi}{4}) + O(1/s)\bigr)$ and $J_1(s) = \sqrt{2/\pi s}\,\bigl(\sin(s-\frac{\pi}{4}) + O(1/s)\bigr)$ [14, §7.21]. Hence $\varrho e^{i\beta} = \sqrt{2/\pi}\,e^{i(s+\pi/4)} + O(1/s)$, and by continuity $H(s) + \frac{\pi}{4} = 2\pi\ell + O(1/s)$ for a fixed integer ℓ. The angle β meets $\frac{\pi}{2} + \pi\mathbb{Z}$ exactly at the zeros of J_1, and always increasingly, since there $\beta' = 1$. So at the n-th positive zero $j_{1,n}$ of J_1 we have $\beta = \frac{\pi}{2} + n\pi$, that is, $H(j_{1,n}) = n\pi - j_{1,n}$. McMahon's expansion $j_{1,n} = (n+\frac14)\pi + O(1/n)$ [14, §15.53] gives $\ell = 0$. □

Lemma 4.3. *Let $0<\delta\le 1$ and $E\ge 1$. There is a constant C_κ, depending only on κ, such that $|\eta(\delta) - H(K(\delta))| \le C_\kappa(\delta^2 + \delta/E)$.*

Proof. On $[0,\delta]$ we use $s = K(t)$ as the variable. Since $\sin 2\varphi = -\sin(2s+2\eta)$, (12) gives

$$\frac{\mathrm{d}\eta}{\mathrm{d}s} = \frac{\dot\varphi - k}{k} = -\frac{a}{k}\sin(2s+2\eta) = F(s,\eta) - r(s)\sin(2s+2\eta),$$

where $F(s,y) = -\frac{1}{2s}\sin(2s+2y)$ and $r = a/k - 1/(2s)$. We first show $|r(s)| \le C_\kappa s/E^2$ for $t\le 1$. Here $E\le k\le E + C_\kappa/E$ and $\dot k = \kappa^2\sinh t\cosh t/k \le C_\kappa t/E$. Write

$$r = \frac{\coth t - 1/t}{2k} + \frac{\kappa^2\sinh t\cosh t}{2wk} + \frac{s - tk(t)}{2tk(t)s}.$$

The first term is at most $t/(6E)$ and the second at most $C_\kappa t/E^3$. For the third, $|s - tk(t)| = \bigl|\int_0^t (k(\tau)-k(t))\,\mathrm{d}\tau\bigr| \le C_\kappa t^3/E$ and $s\ge Et$, so it is at most $C_\kappa t/E^3$. As $t\le s/E$, the bound on r follows.

Let $y(s;\sigma,\xi)$ be the solution of $y' = F(s,y)$ with $y(\sigma) = \xi$, and $Y = \partial_\xi y$. Then

$$Y(s;\sigma,\xi) = \exp\Bigl(-\int_\sigma^s \frac{\cos(2\tau + 2y(\tau))}{\tau}\,\mathrm{d}\tau\Bigr).$$

We claim $Y \le e^9 \max(1, 1/\sigma)$ for $0 < \sigma \le s$. The part of the integral over $[\sigma, 1]$ is at most $\log^+(1/\sigma)$ in modulus. On $[1, \infty)$ put $\theta = 2\tau + 2y$, so that $\theta' = 2 - \sin\theta/\tau \in [1, 3]$ and $|\tau\theta''| \le 4$. Integration by parts,

$$\int_A^B \frac{\cos\theta}{\tau}\,\mathrm{d}\tau = \Big[\frac{\sin\theta}{\tau\theta'}\Big]_A^B + \int_A^B \sin\theta\,\frac{\theta' + \tau\theta''}{(\tau\theta')^2}\,\mathrm{d}\tau,$$

bounds this part by $2 + 7\int_1^\infty \tau^{-2}\,\mathrm{d}\tau = 9$.

Differentiating the flow property shows $\partial_\sigma y(s;\sigma,\xi) = -Y(s;\sigma,\xi)F(\sigma,\xi)$. Hence $Z(\sigma) = y(s;\sigma,\eta(\sigma))$ satisfies $Z'(\sigma) = -Y(s;\sigma,\eta(\sigma))\,r(\sigma)\sin(2\sigma + 2\eta(\sigma))$, which is Alekseev's formula [1]. Integrating from σ_0 to s and comparing $y(s;\sigma_0,\eta(\sigma_0))$ with $H(s) = y(s;\sigma_0,H(\sigma_0))$, we obtain

$$|\eta(s) - H(s)| \le e^9 \max\Big(1, \frac{1}{\sigma_0}\Big)|\eta(\sigma_0) - H(\sigma_0)| + e^9 \int_{\sigma_0}^s \max\Big(1, \frac{1}{\sigma}\Big)|r(\sigma)|\,\mathrm{d}\sigma.$$

By Lemma 3.1 and (16), $\eta(\sigma_0) - H(\sigma_0) = o(\sigma_0)$, so the first term tends to 0 as $\sigma_0 \to 0$. The integrand is at most $C_\kappa(1+\sigma)/E^2$. Taking $s = K(\delta) \le (E + C_\kappa)\delta$ gives the claim. □

Lemma 4.4. *Let $E \ge 8$ and $\delta = E^{-1/3}$. Then $|\eta(\infty) - \eta(\delta)| \le CE^{-2/3}$ with an absolute constant C.*

Proof. For $t \ge \delta$, (13) gives $a \le \frac{1}{2}E^{1/3} + 1 \le E/2 \le k/2$, so $\dot\varphi \ge k/2$. Since $\dot\eta = a\sin 2\varphi$, the same integration by parts as in Lemma 3.2 gives

$$|\eta(\infty) - \eta(\delta)| \le \frac{a(\delta)}{k(\delta)} + \frac{1}{2}\int_\delta^\infty \Big(\frac{2|\dot a|}{k} + \frac{4a\dot k}{k^2} + \frac{4a|\dot a|}{k^2} + \frac{4a^2}{k}\Big)\,\mathrm{d}t.$$

We use $k \ge E$, $k \ge \sqrt{E^2 + b^2e^{2t}}$ with $b = \kappa/2$, hence $\int_0^\infty \mathrm{d}t/k \le E^{-1}\operatorname{arcsinh}(E/b) =: L/E$, together with (13), which gives $a \le \frac{3}{2t}$ and $|\dot a| \le \frac{3}{2t^2}$ on $(0,1]$ and $a \le \frac{6}{5}$, $|\dot a| \le \frac{3}{2}$ on $[1,\infty)$. The boundary term is at most $(\frac{1}{2\delta} + 1)/E$. The four integrals are at most $\frac{3}{2E\delta} + \frac{3L}{E}$, then $\frac{6}{E\delta}$, then $\frac{9}{2E^2\delta^2} + \frac{8L}{E^2}$, and then $\frac{9}{E\delta} + \frac{6L}{E}$. Since $L = O(\log E)$ and $E\delta = E^{2/3}$, the sum is $O(E^{-2/3})$. □

Theorem 4.5. $\Theta(E) = E\log\frac{4E}{\kappa} - E + \frac{\pi}{4} + O\big(E^{-2/3}\big)$ *as* $E \to \infty$.

Proof. Take $\delta = E^{-1/3}$. Then $K(\delta) \ge E\delta = E^{2/3}$, so Lemmas 4.2, 4.3 and 4.4 give

$$\eta(\infty) = H(K(\delta)) + O(\delta^2 + \delta/E) + O(E^{-2/3}) = -\frac{\pi}{4} + O(E^{-2/3}).$$

Inserting this and Lemma 4.1 into (15) proves the theorem. □

Proof of Theorem A. Continuity and monotonicity are Proposition 3.5, the criterion and simplicity are Proposition 3.4 and Lemma 2.1, and the asymptotics is Theorem 4.5, with $4/\kappa = 2/(\pi\lambda^2)$. The counting law follows from Corollary 3.6. If j_+ is the largest integer with $j_+\pi \le \Theta(0)$, Corollary 3.6 gives $\Theta(E_n^+) = (n + j_+)\pi$, and Theorem 4.5 turns this into the stated rule with $n_+ = j_+$. The odd case is the same with half-integers. □

5. The Dirac operator and the zeros of zeta

Let $\lambda = \sqrt{2}$, so $\kappa = 4\pi$. By [6, Prop. 5.3] and [7, Prop. 4.3 and the proof of Thm. 5.1] the spectrum of $2\not{D}$ is $\{\pm 2\sqrt{\mu}\}$ with μ running over the spectrum of the restriction of W_{sa}^+ to even functions supported in $|x| \ge \lambda$. By Section 2 the negative eigenvalues of this restriction are exactly the negative eigenvalues $-(E_n^+)^2$ of W_{sa}^+. The eigenvalues of $2\not{D}$ with positive imaginary part are therefore $2iE_n^+$, and $t_n = 2E_n^+$.

Proof of Theorem B. With $E = t/2$ and $\kappa = 4\pi$ we have $4E/\kappa = t/2\pi$, so by Theorem 4.5 and Stirling's formula for ϑ [12, §4.17],

$$\Theta(t/2) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} + \frac{\pi}{4} + O(t^{-2/3}) = \vartheta(t) + \frac{3\pi}{8} + O(t^{-2/3}).$$

Since $\Theta(t_n/2) = (n + j_+)\pi$, this gives the rule with $n_0 = j_+$. Corollary 3.6 gives $N(T) = N_+(T/2) = \vartheta(T)/\pi + O(1)$, and

$$\frac{\vartheta(T)}{\pi} = \frac{T}{2\pi}\Big(\log\frac{T}{2\pi} - 1\Big) + O(1),$$

which is (2). If (3) held as well, subtracting would give $(\log T)/2\pi = O(1)$, which is absurd.

The Riemann–von Mangoldt formula in the form $N_\zeta(T) = \vartheta(T)/\pi + 1 + S(T)$ [12, §9.3] gives $N(T) - N_\zeta(T) = -S(T) + O(1)$. Selberg showed that $S(T)$ is unbounded in both directions; Tsang [13] proved unconditionally that $S(T) = \Omega_\pm\big((\log T/\log\log T)^{1/3}\big)$. □

Remark 5.1. The term $-(\log E)/2\pi$ in (3) has a simple origin. After the substitution $x = \lambda\cosh t$ and the Liouville transformation, the reduced potential contains $-1/(4\sinh^2 t)$, which is the critical inverse-square singularity at $t = 0$. Counting formulas of Heywood type compare the phase with $\int\sqrt{E^2 + V}$, and with this singularity such an integral diverges logarithmically at $t = 0$; truncating it near $t \approx 1/E$ produces a term of size $\frac{1}{2}\log E$ in the phase, hence $(\log E)/2\pi$ in the count. In the true problem the solution regular at $t = 0$ behaves like $\sqrt{t}\,J_0(Et)$ for $t \ll 1$, whose phase converges to the finite value $\pi/4$ (Lemma 4.2). This is exactly what the Prüfer angle in (10) records: the singular coefficient $\frac{1}{2}\coth t$ in (12) drives the phase from $\pi/2$ to $\pi/2 - \pi/4$ relative to $K(t)$, and nothing more.

Remark 5.2. The rule of Theorem A is uniform in the parity: the union of the even and odd negative spectra is $\{-E^2 : \Theta(E) \in \frac{\pi}{2}\mathbb{Z}\}$. For $\lambda = \sqrt{2}$, the odd eigenvalues correspond to $\vartheta(t) = (m + \frac{1}{8})\pi + O(t^{-2/3})$, halfway between consecutive even ones.

Declarations

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